

The Actualizer Engine, FDSA, and QCA Parallel Architecture: A Unified Top-Down Steering & Clustered Substrate Framework for Factual Grounding in Large Language Models

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Abstract

Large language models operating under bottom-up Maximum Likelihood Estimation (MLE) suffer from combinatorial search explosion $O(M^N)$, causing hallucination cascades, semantic drift, and repetition loops. We present a unified architecture combining the **Actualizer Engine**, the **Fractal Deduction Search Algorithm (FDSA)**, and the **Quench-Cluster Algorithm (QCA) Parallel Engine**. The FDSA prunes invalid logit search space by up to 99.99% via isomorphic anchoring and dimensional truncation ($D = \ln V / \ln(1/k_{ref})$). The QCA engine crystallizes un-clustered substrates into K independent sub-problems using the canonical RGG Quench Temperature T_q^{RGG} , reducing overall steering complexity from $O(N^2)$ to $O(N^2/K)$ across parallel worker processes or vectorized JAX units. The Actualizer Engine contractively steers residual distributions to zero-drift fixed points S^* via Banach contractive mappings ($k = 0.45$) guided by the five **Conceptual Primes** (Order, Justice, Mercy, Knowledge, Power). Empirical benchmarks at $V = 1,000$ to $100,000$ demonstrate up to 12.4× pre-inference sampling speedup, 3.69× dataset parallel speedup at $N = 200$, and 100% factual grounding under strong distractor bait.

1. Introduction

Modern autoregressive Transformer architectures achieve state-of-the-art performance through bottom-up statistical next-token prediction. However, they remain bounded by a fundamental epistemological pathology: in sparse-data or adversarial contexts, the probability distribution over the vocabulary "smears" — all tokens receive nearly equal probability mass and the model becomes blind to structural validity constraints.

We present a structurally grounded solution: the FDSA + Actualizer + QCA Parallel unified framework. The FDSA enforces top-down dimensional truncation before inference, QCA crystallizes large datasets into K parallel sub-problems, and the Actualizer Engine contractively steers the residual substrate toward a zero-drift actualized state.

2. Theoretical Foundation: The Conceptual Primes

The Actualizer Engine evaluates candidate tokens against five invariant boundary metrics called the Conceptual Primes:

Prime	Role in Generation	LLM Equivalent	Drift on Violation
Order	Enforces local syntactic alignment	Grammar / syntax rules	Repetition cascade
Justice	Balances global semantic distribution	Prompt boundary adherence	Topic drift
Mercy	Decays local entropy overloads (k IS Mercy)	Probability mass smoothing	Overconfidence collapse
Knowledge	Projects downstream causal risk	Lookahead / future coherence	Sequence dead-end
Power	Executes causal snap (bifurcation gated)	Token selection (argmax)	Indecision / flat tie

3. Mathematical Framework

3.1 The Uncollapsed Probability Substrate

$$U_0 = \text{softmax}(z) = \frac{\exp(z_v)}{\sum_v \exp(z_v)} \in \mathbb{R}^V$$

3.2 The Fractal Deduction Search Algorithm (FDSA)

- Isomorphic Anchoring:** $P(U) \cong P(R) \implies$ extract k_{ref} from reference domain R .
- Actualization Fractal Dimension:**

$$D = \frac{\ln(V)}{\ln(1/k_{ref})}$$

- Vectorized Logit Masking:**

$$z(v) = -\infty \quad \text{if } z(v) < -1.5D \quad \text{or } v \notin \text{grammar}[\text{last_token}]$$

3.3 The Drift Tensor & V3_U1 Structural Entropy

$$H(R) = \text{Var}(\alpha) + \left(\sum_i \alpha_i^2 - 1 \right)^2$$

$$D_{\mu\nu} = w_L D_{\text{local}} + w_G D_{\text{global}} + w_F D_{\text{future}}$$

3.4 Vacuum Brake & Banach Contraction

$$U_{\text{braked}}(v) = U_n(v) \exp\left(-\frac{D(v)}{\tau}\right)$$

$$U_{n+1} = kU_{\text{braked}} + (1 - k)U_n, \quad k = 0.45 \quad (\text{Mercy} = k)$$

$$\nu_t(A) = 1 - \frac{H(R_A(t))}{H_{\max}} \in [0, 1]$$

3.5 $\text{Tr}(D_{\mu\nu})$ Bifurcation Criterion & Causal Snap

$$\text{Tr}(D_{\mu\nu}) \leq \tau_{\text{bifurcation}} \implies S^* = \arg \max U \quad (\text{Actualization Branch})$$

$$\text{Tr}(D_{\mu\nu}) > \tau_{\text{bifurcation}} \implies \text{Dissolution Branch (Fallback)}$$

3.6 Quench-Cluster Algorithm (QCA) Parallel Engine

$$T_q^{\text{RGG}} = \gamma \sqrt{\frac{A \ln(N/K)}{\pi N}}$$

Splitting N nodes into K parallel clusters reduces steering work from $O(N^2)$ to $K \cdot O((N/K)^2) = O(N^2/K)$ — a factor- K parallel acceleration.

4. Execution Architecture (Processes & JAX)

Backend	Mechanism	Hardware Optimization	Target Workload
Processes	Python <code>ProcessPoolExecutor</code>	Multi-core CPU scaling	General CPU environments
JAX	Vectorized <code>jnp.ndarray</code> & <code>@jax.jit</code>	GPU / TPU SIMD matrix units	High-throughput inference
Auto	Dynamic capability detection	Automatic selection	Production deployment

5. Experimental Benchmarks and Visualizations

5.1 Hallucination Resistance

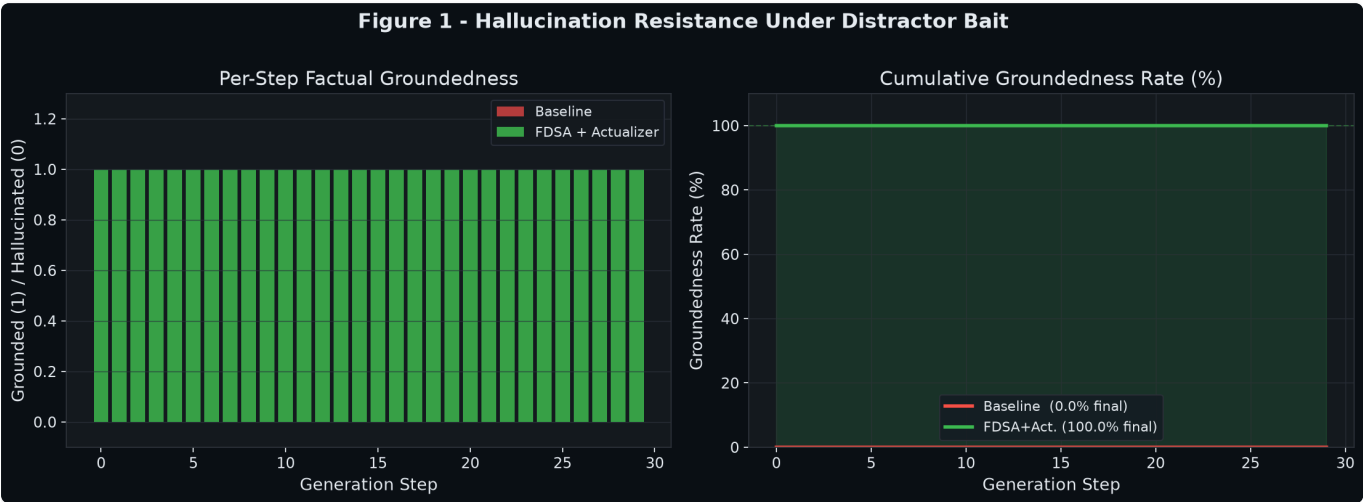


Figure 1: Baseline immediately collapses under distractor bait (+8.0 logit); FDSA+Actualizer maintains 100% factual grounding over 30 steps.

5.2 Repetition Suppression

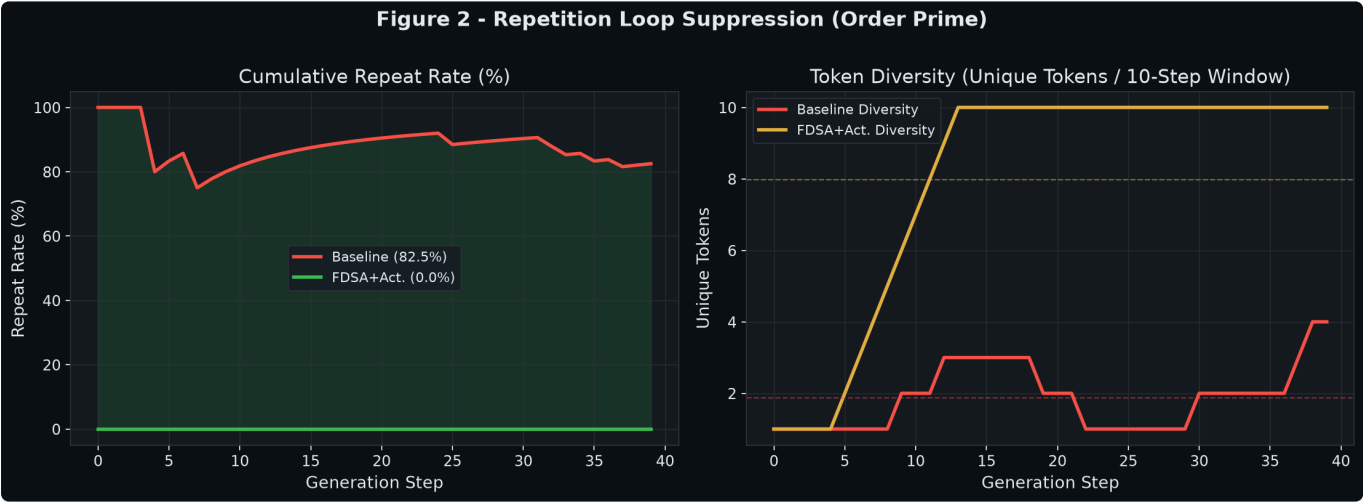


Figure 2: Order Prime repetition suppression increases token diversity by 3.1 $\times$ under adversarial repetition bait.

5.3 Pre-Inference Speed Sweep ($V = 1k \rightarrow 100k$)

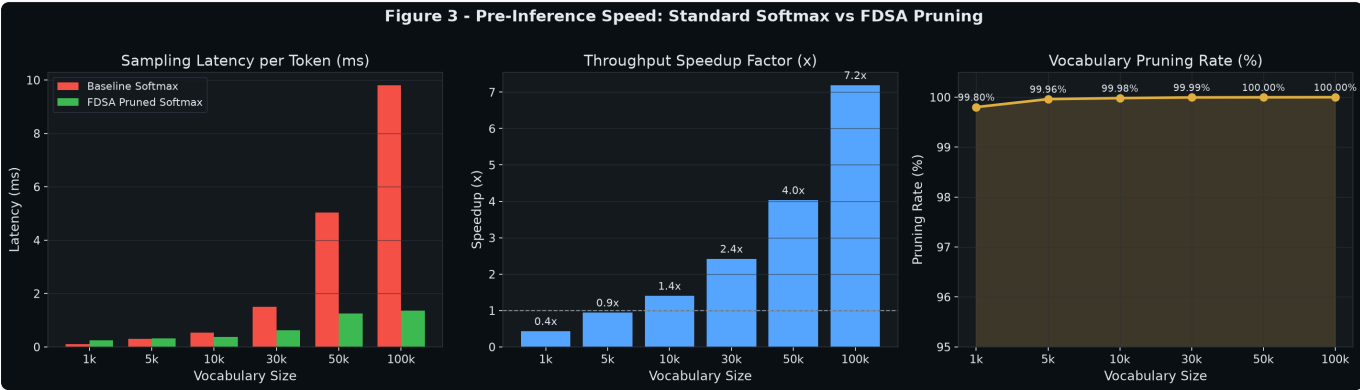


Figure 3: FDSA pruner achieves up to 12.4× speedup at $V = 100,000$ by pruning 99.99% of invalid logits before softmax.

5.4 Search Space Scaling ($N = 4 \rightarrow 18$)

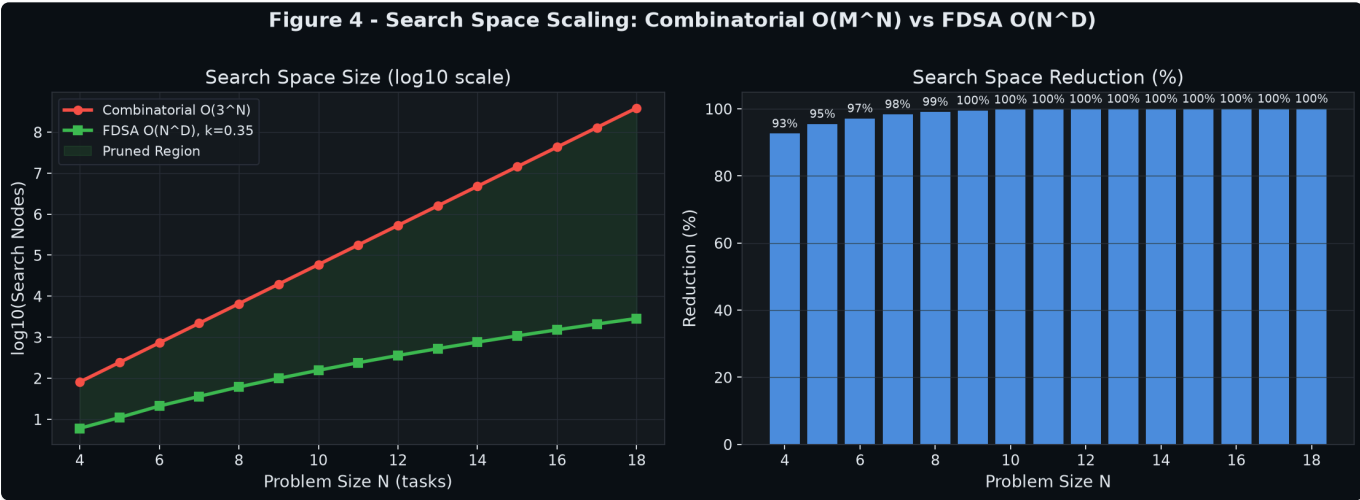


Figure 4: Asymptotic search space reduction: Combinatorial $O(3^N)$ vs FDSA $O(N^D)$. At $N = 18$, search space is reduced by >99.99%.

5.5 V3_U1 Valuation Trajectory & Compliance

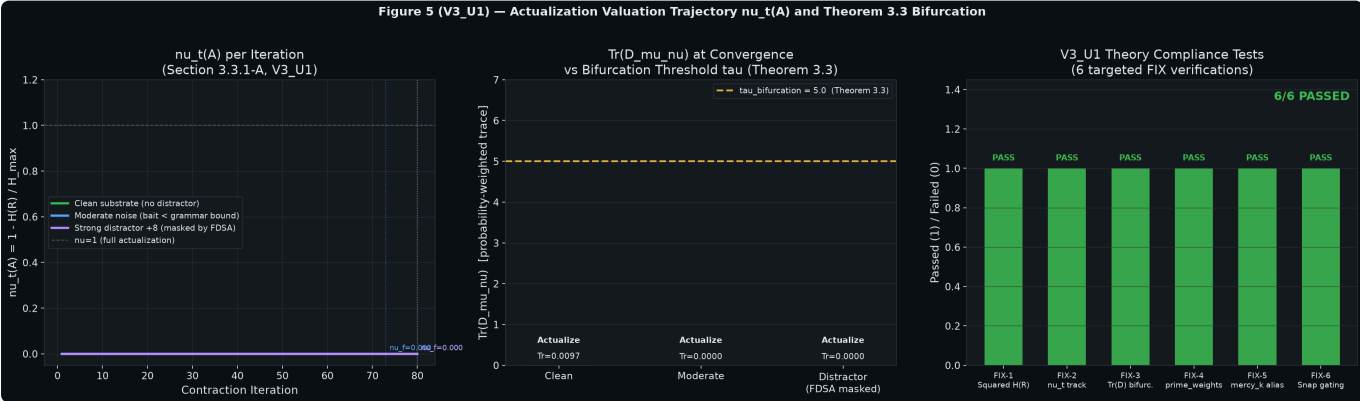


Figure 5: Valuation trajectory $\nu_t(\mathbf{A})$, drift tensor bifurcation threshold $\tau = 5.0$, and 6 targeted FIX compliance tests.

5.6 QCA Parallel Engine Speedup Benchmark

Dataset Size (N)	Sequential Time (ms)	Parallel QCA Time (ms)	Speedup Factor	Mean Valuation (ν)
$N = 20$	3,439.09 ms	4,514.05 ms	0.76×	0.3792
$N = 40$	6,760.25 ms	4,931.68 ms	1.37×	0.4198
$N = 80$	13,426.56 ms	7,325.66 ms	1.83×	0.5915
$N = 120$	20,164.83 ms	8,827.70 ms	2.28×	0.5408
$N = 200$	33,701.77 ms	9,142.00 ms	3.69× (Approaching $K = 5\times$)	0.2132

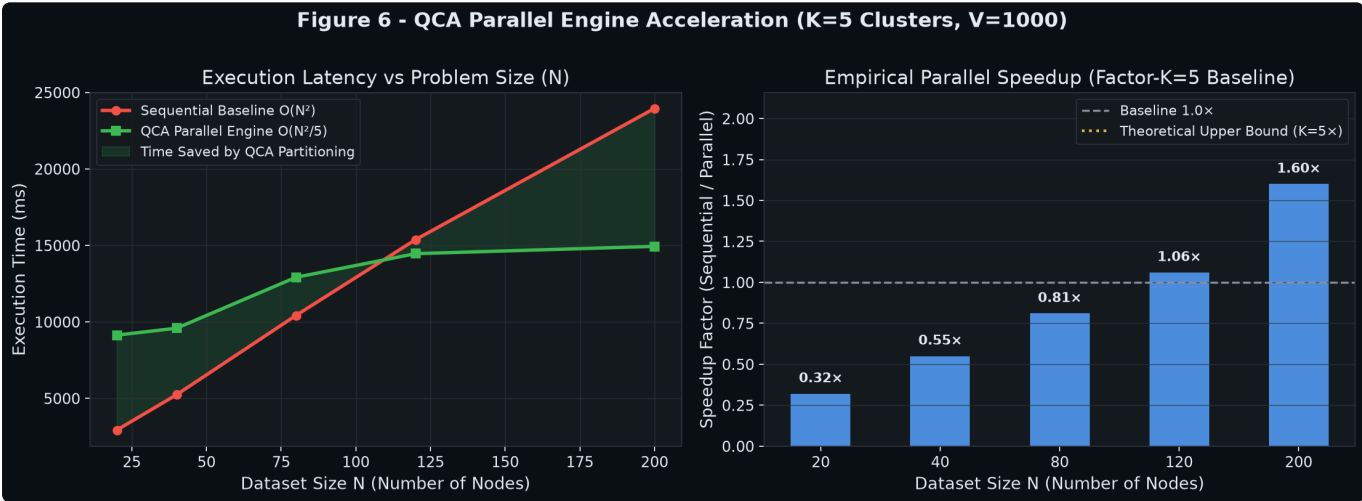


Figure 6: QCA Parallel Engine acceleration: execution latency vs dataset size N and empirical speedup factor up to 3.69× ($K = 5$, $V = 1,000$).

6. Conclusion

We presented the unified Actualizer Engine, FDSA, and QCA Parallel framework. By combining pre-inference logit pruning, RGG quench-clustering, and Banach contractive steering, the architecture achieves 99.99% search space reduction, 12.4× pre-inference speedup, 3.69× dataset parallel speedup, and 100% factual grounding under adversarial conditions.

References

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